# SOFTWARE ENGINEERING – LAB 03

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For the Municipal Infrastructure Reporting App, I would use a layered architecture. The architecture divides the system into three logical layers: presentation layer, business/application layer and data layer. The presentation layer contains citizen and engineer interface through which citizens submit infrastructure damage reports and municipal engineers view and update assigned reports. The business or application layer contains the core processing components: report management, location and GIS, department routing, notification service, and offline sync. The data layer contains the municipal database for storing reports, users, locations, statuses, and departmental assignments.

**Why a Layered Architecture**

**Reason 1: Separation of responsibilities**

Layered architecture separates user interaction from the application's core processing. The report management component can validate and track reports without depending directly on the mobile/web interface. Similarly, location and GIS, Department routing and notification services perform specialized functions through defined interfaces. This makes the individual components easier to maintain or modify without affecting unrelated parts of the system.

**Reason 2: Supports the municipal reporting workflow**

The architecture maps naturally to the application's required workflow. A citizen submits a report through the presentation layer; report management validates it and communicates with location and GIS to obtain the incident location and department routing to identify the responsible municipal division. The report is then stored in the municipal database, and the notification service informs the citizen about submission and subsequent status changes. The offline sync component supports report drafting and synchronization when connectivity is unavailable.

**Security Advantage**

The layered architecture or structure prevents the user interface from directly accessing the municipal database. Requests pass through the business or application layer, where validation and authorization can be applied before sensitive citizen, report and location information is accessed. This provides a controlled boundary between external users and stored municipal data.